

VERITAS UNIVERSITY, ABUJA

Course Title: Computer Science Innovation and New
Technologies

Course Code: CSC 322

**Explainable AI for Credit Risk Scoring Using Machine
Learning**

Group 5 Report

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Date: 25th May, 2026

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Introduction

Credit risk scoring is an important process used by banks and financial institutions to determine whether a customer is capable of repaying a loan. Traditional methods of credit scoring are often manual, slow, and difficult to interpret. Artificial Intelligence (AI) and Machine Learning (ML) can improve this process by automatically learning patterns from financial data and making accurate predictions.

In this project, an Explainable AI Credit Risk Scoring System was developed using the German Credit Dataset. The system predicts whether a customer represents a good or bad credit risk based on financial information such as loan amount, employment status, savings, and account status.

To improve transparency and trust in the model, Explainable AI techniques such as SHAP and LIME were explored. A Streamlit web application was also developed to allow users interact with the prediction system easily.

Objectives

- To develop a machine learning model for credit risk prediction.
- To use the German Credit Dataset for training and testing.
- To apply Explainable AI techniques for model interpretation.
- To deploy the model using Streamlit.
- To improve transparency and understanding of AI predictions.

Dataset Description

The dataset used for this project is the German Credit Dataset obtained from Kaggle. The dataset contains information about customers and their financial background.

The dataset contains 1000 records with 20 features including credit amount, loan duration, employment years, savings status, housing information, age, and account status. The target column classifies customers as either good credit risk or bad credit risk.

The target variable classifies customers into Good Credit Risk & Bad Credit Risk.

Methodology

1. Data Preprocessing

The dataset was cleaned and categorical features were converted into numerical format using Label Encoding.

2. Feature Selection

Input features were separated from the target variable.

3. Data Splitting

The dataset was divided into training and testing sets using `train_test_split()`.

4. Model Training

A Random Forest Classifier from Scikit-learn was used to train the model.

5. Model Evaluation

The model was tested using accuracy score evaluation.

6. Explainability

SHAP and LIME techniques were explored to explain predictions and improve transparency.

SHAP identified globally important features affecting credit risk prediction.

LIME explained individual predictions and showed which financial factors contributed positively or negatively to the result.

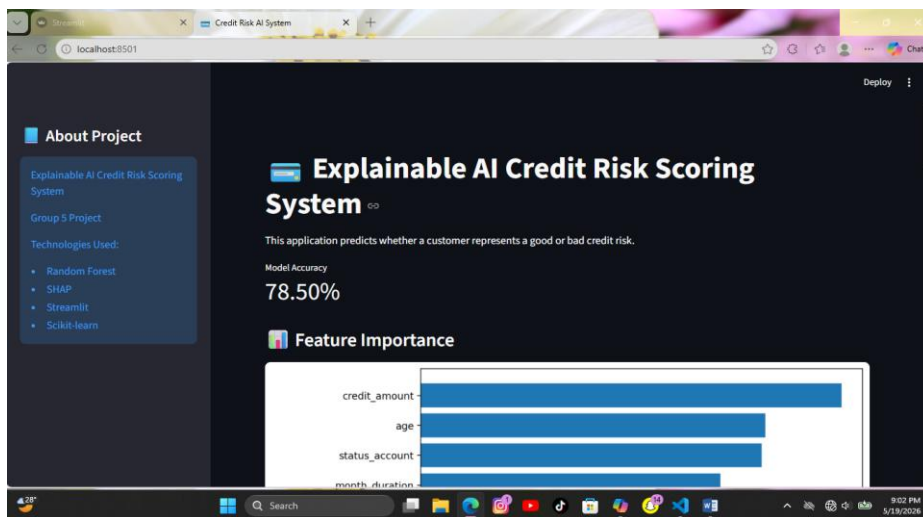
These explanations improved transparency and understanding of the AI system.

Results and Visuals

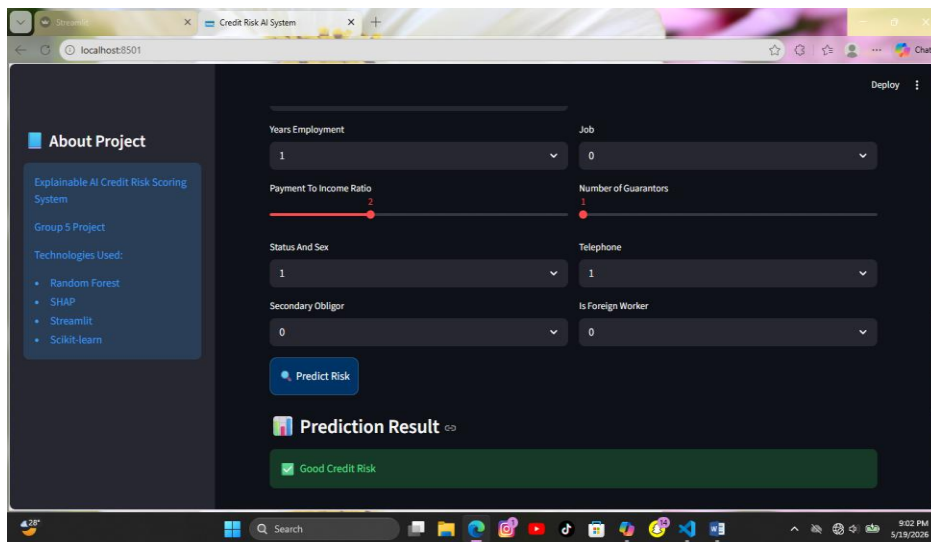
The Random Forest model achieved an accuracy score of 78.5%, showing good performance in predicting customer credit risk.

The model was able to classify customers into good and bad credit risk categories based on their financial information.

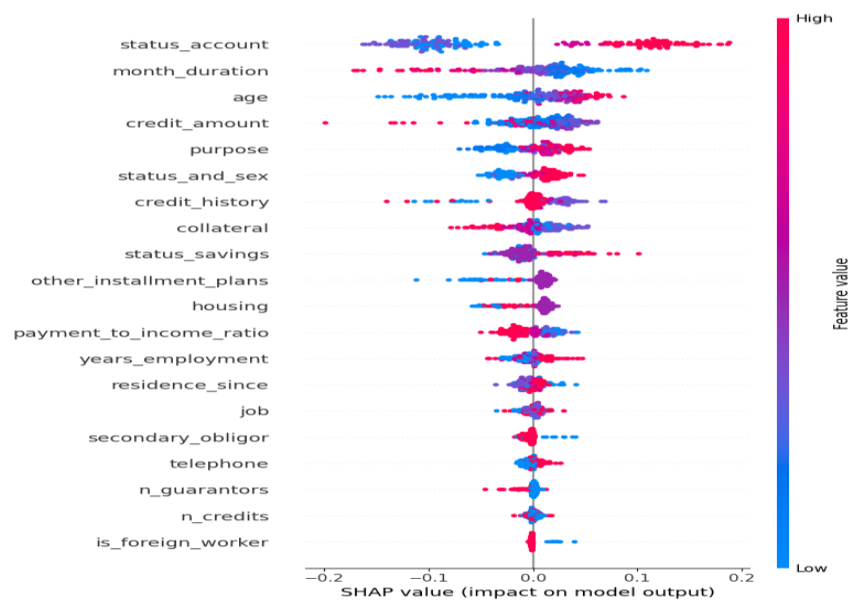
1. Streamlit App- Credit Risk Scoring System



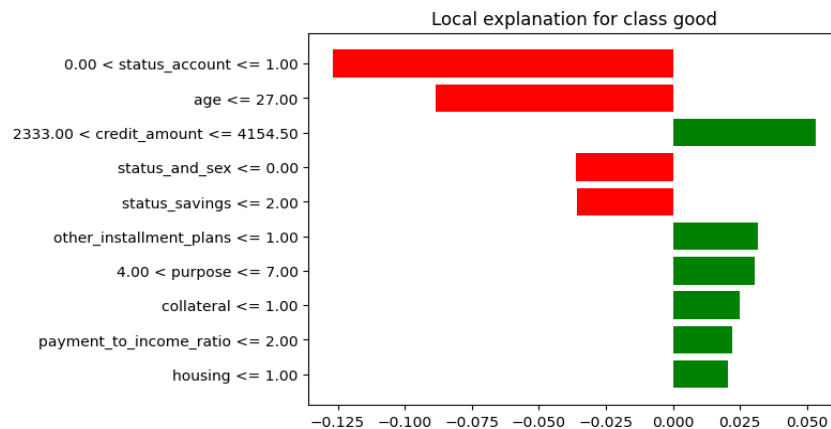
2. Prediction Result



3. SHAP Visualization



4. LIME Visualization



Fairness and Bias

One important challenge in AI systems is fairness and bias. Some features such as gender and employment status may introduce bias into predictions if not handled carefully.

The project considered explainability techniques to improve transparency and help identify possible bias in the model decisions.

Deployment

The trained machine learning model was deployed using Streamlit to create an interactive web application.

Users can enter customer financial details through the interface and receive instant credit risk predictions.

The application improves usability and demonstrates practical implementation of Explainable AI systems.

Conclusion

This project successfully developed an Explainable AI Credit Risk Scoring System using machine learning techniques. The Random Forest model achieved good prediction accuracy and Explainable AI methods improved understanding of the predictions.

The Streamlit application provided a simple and interactive platform for users to test the model. The project demonstrates how AI can support financial decision-making while maintaining transparency and interpretability.

References

1. Scikit-learn Documentation – <https://scikit-learn.org>
2. SHAP Documentation – <https://shap.readthedocs.io>
3. Streamlit Documentation – <https://streamlit.io>
4. German Credit Dataset – Kaggle